Lecture 2 - Gauge theories on a lattice

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Abstract

In this lecture we begin by discussing the fate of spacetime symmetries on a discrete lattice. We then briefly remind ourselves of the continuum abelian gauge theory and gauge transformations. Following this we see that demanding gauge invariance for the scalar field theory (on the lattice) gives new structure in constrast to what is done in the continuum. We then discuss the basics of lattice gauge theories starting from defining links and plaquettes, building the Wilson gauge action. If time permits, we then end with explaining how to fix a gauge on the lattice using maximal trees, finally introducing some important observables like the Wilson and Polyakov loops. This lecture is based on many references including [1, 2, 3, 4].

1 Overview: Symmetries on the Lattice

As we have seen previously, the discrete spacetime lattice — with properly defined lattice fields and observables — provides us with a powerful tool to compute finite quantities whose correct continuum extrapolation leads to expected results. However, as one might expect, the discrete lattice explicitly breaks certain continuum symmetries, which must be recovered in the continuum limit.

We can broadly divide these symmetries into two classes: spacetime and internal (field). When considering spacetime symmetries, one should keep in mind that we are working in *Euclidean* space - obtained by Wick rotation from the Minkowski formulation - so that the relevant continuum symmetry group is $\mathrm{SO}(4)$ rather than the Lorentz group $\mathrm{SO}(3,1)$. This is a crucial simplification: Lorentz invariance, involving non-compact boosts, is *qualitatively* broken on any discrete lattice, a boosted time-slice generically misses all lattice sites, and there is no controlled sense in which the lattice approximates boost symmetry. By contrast, the full rotation group $\mathrm{SO}(4)$ is broken only to the discrete hypercubic group $H(4)$, and the resulting symmetry-violating operators are irrelevant in the renormalization group sense - suppressed by powers of the lattice spacing a - so that $\mathrm{SO}(4)$ invariance is systematically recovered in the continuum limit.

See David Tong's lecture notes[5] on Lattice Field Theory for a insightful treatment on broken Lorentz invariance on the (Minkowski) lattice.

This discussion on $H(4)$ symmetry is taken from Guy Moore's lecture[6] on LFT and is explained there in great detail.

Internal symmetries, on the other hand, need not be affected by the discretization of spacetime at all. A simple example is the $\phi(x) \to -\phi(x)$ symmetry of the free scalar field, which holds identically on the lattice since it acts pointwise and does not involve derivatives or transport between sites. In this lecture, we will consider much more tricky example: what happens when we demand *local gauge invariance* on the lattice.

Note: In this lecture we will restrict ourselves to the (mathematically simpler) case of abelian gauge theories, which physically describes only electromagnetic interactions. To study QCD one must go to non-abelian gauge theories, which lead to very distinct phenomenology best covered in a proper QFT course. As an invitation: non-abelian theories describe colored gluons, and this "color" charge leads to self-interactions between gluons - something that is absent for photons. This self-interaction gives rise to phenomena such as asymptotic freedom, meaning the coupling becomes weak at short distances, which turns out to be essential for taking the continuum limit of QCD on the lattice. More on the continuum limit in a later lecture.

2 Continuum Abelian Gauge Theory: A Recap

Before discussing how to make our theory locally gauge invariant, it is good to remind ourselves what gauge fields are and what action describes them - in the continuum. One should recall that the Standard Model of particle physics is made up of matter fields (like scalar and fermions) and force fields (photons and gluons). Quantum electrodynamics (QED) describes the interaction of fermions (electrons, muons so on) with photons which are described by an *abelian gauge field theory*. While quantum chromodynamics (QCD) describes the dynamics of quarks and gluons, where the gluons are described by *non-abelian gauge fields*.

Due to the mathematical simplicity of abelian gauge fields, we will restrict ourselves in this lecture to studying those. Euclidean abelian gauge field theory is described by the following *Yang-Mills* action,

You may notice that the Euclidean Yang-Mills action comes with a positive sign as opposed to the Minkowski version.

$$S[A] = \frac{1}{4}\int d^4x\; F_{\mu\nu}(x)F_{\mu\nu}(x)\,, \tag{1}$$

where $F_{\mu\nu}(x) = \partial_\mu A_\nu(x) - \partial_\nu A_\mu(x)$ is the anti-symmetric field strength tensor and $A_\mu(x)$ is the vector potential. The field strength tensor $F_{\mu\nu}(x)$ has the property that it is invariant under the follwing *gauge* transformation,

$$A'_\mu(x) = \Omega(x)\,A_\mu(x)\,\Omega^\dagger(x) - i\Omega(x)\,\partial_\mu\Omega^\dagger(x)\,, \tag{2}$$

when $\Omega(x)$ are elements in the abelian group $\mathcal{G} = U(1)$. These are then just phase factors and commute with each other, $\Omega(x)\Lambda(x) = \Lambda(x)\Omega(x)$ and have the property that $\Omega^\dagger(x) = \Omega^{-1}(x)$. Choosing such a phase factor to be $\Omega(x) = e^{i\alpha(x)}$, and using the abelian property (which allows us to move these phase factors past each other), Eq. (2) becomes,

$$A'_\mu(x) = A_\mu(x) + \partial_\mu\alpha(x)\,. \tag{3}$$

The transformation represented by Eq. (3) is what is usually known as *local gauge transformation* and you might remember this from your course in Electrodynamics, as these transformations leave physical observables like electric and magnetic fields invariant. As can also be easily checked, this transformation also leaves the action $S[A]$ invariant.

There are nice discussions on this in Guy Moore's lecture notes[6] for non-abelian gauge theories. For the abelian side of things, there a few concrete integrals on page 97 in Jan Smit's LFT book[2].

This gauge invariance introduces a problem when we try to naively construct the path inetgral for the gauge theory. To see this, notice that for every configuration $A_\mu(x)$, there exist infinitely many gauge-equivalent configurations related by Eq. (3)

that give identical physics - the same action, the same observables. These gauge-equivalent configurations form an *orbit of the gauge group* $\mathcal{G}$. The measure $\mathcal{D}A_\mu$ integrates over all configurations, including all copies within each orbit, so it factorizes as

$$\mathcal{D}A_\mu = \mathcal{D}A_\mu^{\text{phys}} \cdot \mathcal{D}\alpha, \qquad \mathcal{D}\alpha = \prod_x d\alpha(x)\,, \tag{4}$$

where $\mathcal{D}A_\mu^{\text{phys}}$ is the measure over gauge-inequivalent configurations and $\mathcal{D}\alpha$ is the measure over the gauge group. Since the action is gauge invariant, the integral over $\mathcal{D}\alpha$ yields only the volume of the gauge group,

$$Z = \text{Vol}(\mathcal{G}) \int \mathcal{D}A_\mu^{\text{phys}}\, e^{-S[A]}\,, \tag{5}$$

which diverges since $\alpha(x) \in \mathbb{R}$ at every spacetime point, making $\text{Vol}(\mathcal{G}) = \infty$. This infinite overcounting must be dealt with - in the continuum this is achieved via the Faddeev-Popov procedure[1].

We will see below that on the lattice, gauge invariance is built in exactly and the problem is resolved differently: the link variables $U \in SU(N)$ become the fundamental degrees of freedom instead of the vector potential and take values in a compact group, so $\text{Vol}(\mathcal{G})$ is finite and the path integral is naturally regulated without the need for gauge fixing.

3 Local Gauge Invariance in the Lattice Free Scalar Field Theory

In this section we will begin by considering some internal symmetries of the scalar field theory given by a complex scalar field $\phi(x) = (\phi_1(x) + i\phi_2(x))/\sqrt{2}$, whose (Euclidean) action is given by,

$$S_E[\phi] = \int d^4x\, \left\{\partial_\mu\phi^*(x)\partial_\mu\phi(x) + M^2|\phi|^2(x)\right\}\,. \tag{6}$$

Looking at the action, it is easy to see that, a rotation by a global phase $\phi(x) \to e^{i\alpha}\phi(x)$, leaves the action invariant.

However, if one wants to couple the free scalar field with with gauge fields, then to have a consistent theory, the entire action must respect local gauge invariance as discussed in the previous section . The way gauge invariance acts on matter fields is by applying local, space dependent phases to the fields: $\phi(x) \to e^{i\alpha(x)}\phi(x)$. One immediately sees that under this transformation, the kinetic term of the free scalar field is not invariant and transforms as,

You can learn more about continuum scalar quantum electrodynamics in Chapter 9 of the book by M. Schwartz[4].

$$\partial_\mu\phi'(x) = e^{i\alpha(x)}\left(\partial_\mu\phi(x) + \partial_\mu\alpha(x)\right)\,. \tag{7}$$

The mass term on the other hand is unchanged under the transformation. The procedure to make the full action gauge invariant is by modifying the derivative

[1] This is something you learn in your QFT or Advanced QFT courses

term from $\partial_\mu \to D_\mu$, where $D_\mu = \partial_\mu + iA_\mu$. Assume that the gauge transformation is given by $A_\mu(x) \to A_\mu(x) + \partial_\mu \lambda(x)$, then D_μ transforms as,

$$\begin{aligned} D_\mu \phi(x) &\to D'_\mu \phi'(x) \\ &= \left(\partial_\mu + iA_\mu + i\partial_\mu \lambda(x)\right) e^{i\alpha(x)} \phi(x) \\ &= e^{i\alpha(x)} \left(\partial_\mu + iA_\mu + i\partial_\mu \lambda(x) + i\partial_\mu \alpha(x)\right) \phi(x) \,. \end{aligned} \tag{8}$$

In order to recover gauge invariance - we need to choose $\lambda(x) = -\alpha(x)$ giving us the following rules,

$$\begin{aligned} A_\mu &\longrightarrow A_\mu(x) + \partial_\mu \lambda(x) \,, \\ \phi(x) &\longrightarrow e^{-i\lambda(x)} \phi(x) \,, \\ D_\mu \phi(x) &\longrightarrow e^{-i\lambda(x)} D_\mu \phi(x). \end{aligned} \tag{9}$$

In the continuum, this would result in the following gauge invariant action,

$$S_E[\phi, A] = \int d^4x \; D_\mu \phi^*(x) D_\mu \phi(x) + M^2 \phi^*(x)\phi(x) + \frac{1}{4} F_{\mu\nu}(x) F_{\mu\nu}(x) \,, \tag{10}$$

with the resulting path inetgral,

$$Z = \int \mathcal{D}A \mathcal{D}\phi \; e^{-S_E[\phi, A]} \,. \tag{11}$$

In order to construct a lattice version of the gauge invariant action for the interacting scalar and gauge field theory, we need to consider (i) what happens to the lattice derivative and more importantly, (ii) what observable should we choose to represent the gauge degrees of freedom on the lattice.

To begin the discussion, we should look at Eq. (9) again and realise that the second transformation should also hold on the lattice the phase factor just becomes a function of lattice sites giving,

$$\begin{aligned} \phi'(na) &= e^{-i\lambda(na)} \phi(na) \\ \phi'^*(na) &= \phi^*(na) e^{i\lambda(na)} \,. \end{aligned} \tag{12}$$

For completeness, the dimensionless version of these equations look like,

$$\begin{aligned} \hat{\phi}'_n &= e^{-i\lambda_n} \hat{\phi}_n \\ \hat{\phi}'^*_n &= \hat{\phi}^*_n e^{i\lambda_n} \,. \end{aligned} \tag{13}$$

Considering the (dimensionless) lattice action from the previous lecture - now with a complex scalar field,

$$S_E = -\sum_{n,\hat{\mu}} \left(\hat{\phi}^*_n \hat{\phi}_{n+\hat{\mu}} + \hat{\phi}^*_{n+\hat{\mu}} \hat{\phi}_n \right) + \left(8 + \hat{M}^2 \right) \sum_n \hat{\phi}^*_n \hat{\phi}_n \,, \tag{14}$$

it is easy to see that under a gauge trasnformation, only the *hopping* term of the action transforms non-trivially. We further use the notation of group elements $\Omega(n) = e^{-i\lambda_n}$ to see,

$$\sum_{n,\hat{\mu}} \hat{\phi}^*_n \hat{\phi}_{n+\hat{\mu}} \to \sum_{n,\hat{\mu}} \hat{\phi}^*_n \Omega^\dagger(n) \Omega(n+\hat{\mu}) \hat{\phi}_{n+\hat{\mu}} \,. \tag{15}$$

It should be clear that $\Omega^\dagger(n)\Omega(n+\hat{\mu}) \neq 1$. This leads us to consider the parallel transporter or a Wilson line, which for an abelian gauge theory is simply,

$$U(x,y) = e^{ie\int_x^y \sum_\mu dz_\mu A_\mu(z)} \,. \tag{16}$$

The physical relevance of this quantity is that it tells us how to *transport* a field in gauge space from one spacetime point to another. Moreover, under a gauge trasnformation, it transforms like,

$$U(x,y) \to \Omega(x)U(x,y)\Omega^\dagger(y)\,. \tag{17}$$

This gives us a hint on how to construct the lattice hopping term, such that gauge invariance is preserved. The quantity one should consider is $\phi^*(x)U(x,y)\phi(y)$, which remains invariant under a gauge transformation,

$$\begin{aligned}\phi^*(x)U(x,y)\phi(y) &\to \phi^*(x)\Omega(x)\Omega^\dagger(x)U(x,y)\Omega^\dagger(y)\Omega(y)\phi(y) \\ &= \phi^*(x)U(x,y)\phi(y)\end{aligned} \tag{18}$$

Hence, the gauge invariant action for the complex scalar field is given by,

$$S_E = -\sum_{n,\hat{\mu}} \hat{\phi}_n^* U_{n,n+\hat{\mu}} \hat{\phi}_{n+\hat{\mu}} + \hat{\phi}_{n+\hat{\mu}}^* U_{n+\hat{\mu},n} \hat{\phi}_n + \left(8 + \hat{M}^2\right) \sum_n \hat{\phi}_n^* \hat{\phi}_n \,. \tag{19}$$

In the next section we will consider how these Wilson lines look on the lattice and how to construct the action for the abelian lattice gauge theory.

From now onward we will only insert factors of the lattice spacing a when it is non-trivial, meaning when there is a an expression which is not directly a length - so $x = na$ will almost always appear as n.

4 Gauge Fields on the Lattice: Links, Plaquettes and the Wilson Action

In lattice gauge theory, the fundamental gauge degrees of freedom are not the continuum vector potentials $A_\mu(x)$, but rather group-valued variables associated with the links and plaquettes of the lattice. These variables encode the parallel transport of fields and the curvature (field strength) in a gauge-invariant way.

Link Variables

On a hypercubic lattice, each site is labeled by an integer vector n and each direction by $\hat{\mu}$. The link variable $U_{n,n+\hat{\mu}}$ is an element of the gauge group and is associated with the oriented link from site n to site $n+\hat{\mu}$. For an abelian theory ($U(1)$),

$$U_{n,n+\hat{\mu}} = e^{iaeA_\mu(n)} \tag{20}$$

where a is the lattice spacing and e is the gauge coupling (keeping with the notation of Eq. (16)). As we have already seen, link variable act as parallel transporters for matter fields between neighboring sites. A pictorial representation of links can be seen in Fig. 1, where the link (blue) and its conjugate (red) are shown. Notice that since the links are directional, it makes sense to choose a convention for positve

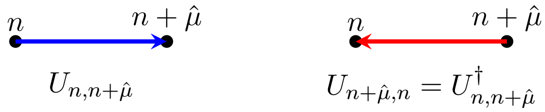


Figure 1: Links: fundamental gauge degrees of freedom on the lattice

directions. For the rest of this lecture series we will choose up and right to be the links, with down and left representing the conjuagtes. This is why the links are colored in the figure.

Before we move on to the next section, we will also adopt the following definition to represent a link (this is just following the conventions of the lattice community). In these conventions, $U_\mu(n) \equiv U_{n,n+\hat{\mu}}$ and $U^\dagger_\mu(n) = U^{-1}_\mu(n) = U_{-\mu}(n+\hat{\mu}) \equiv U_{n+\hat{\mu},n}$.

Plaquette Variables

The smallest nontrivial closed loop on the lattice is the plaquette, a square formed by four links in the μ-ν plane. The plaquette variable at site n is defined as

$$U_{\mu\nu}(n) = U_\mu(n) U_\nu(n+\hat{\mu}) U^\dagger_\mu(n+\hat{\nu}) U^\dagger_\nu(n) \tag{21}$$

For non-abelian gauge theories like $SU(N)$, it is a product of matrices and one needs to take a trace over the product to recover gauge invariance.

For $U(1)$, this is just the product of four phases around the plaquette, shown in Fig. 2. The plaquette is a gauge invariant observable and can be shown to be related to the continuum field strength tensor as follows,

This is part of an exercise in the next sheet.

$$U_{\mu\nu}(n) \approx e^{ia^2 F_{\mu\nu}(n)}\,, \tag{22}$$

where the lattice field strength tensor is given in terms of differences of the vector potential,

$$F_{\mu\nu}(n) = \frac{1}{a}\left[A_\nu(n+\hat{\mu}) - A_\nu(n) - (A_\mu(n+\hat{\nu}) - A_\mu(n))\right]\,. \tag{23}$$

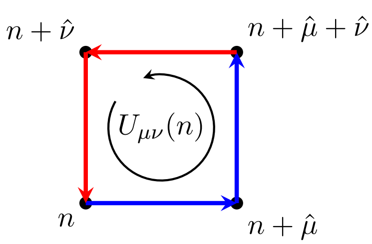
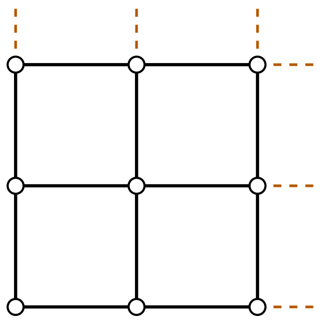


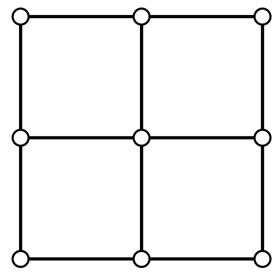
Figure 2: Plaquette: smallest gauge-invariant quantity constructed from links

Boundary conditions & Degrees of Freedom

In lattice field theory simulations one typically has many choices for setting boundary conditions on the lattice. The most commonly used one is *periodic* boundary conditions (PBC), which identify opposite edges of the lattice and preserve translational invariance. The other common choice is *open* boundary conditions (OBC), where the fields at the boundary are left free with no identification imposed. In the infinite volume limit both boundary conditions become equivalent, though the choice of boundary conditions can become important when studying topological properties of the theory. We will see below how the number of links and hence plaquettes differs between the two.



(a) Periodic boundary conditions



(b) Open boundary conditions

Figure 3: Two examples of boundary conditions on the lattice

Looking at Fig. 3, in the left panel the dashed arrows indicate links that wrap around and re-enter the lattice from the opposite edge, realising the periodic identification. In the right panel, under OBC, the links terminate at the boundary with no periodic partner - in two dimensions there is only one link per boundary site (and $D-1$ links in D dimensions). Note that one corner site has no outgoing open links at all, as it sits at the intersection of two boundaries. This difference in boundary structure leads to a different counting of degrees of freedom in the two cases.

On a lattice with D-dimensions and N-sites per dimension, we have the following counting of links and plaquettes for the two boundary conditions described,[2]

$$\text{PBC:} \quad D \cdot N^D \text{ links}, \qquad \frac{D(D-1)}{2} N^D \text{ plaquettes} \tag{24}$$

$$\text{OBC:} \quad D \cdot N^{D-1}(N-1) \text{ links}, \qquad \frac{D(D-1)}{2}(N-1)^2 N^{D-2} \text{ plaquettes} \tag{25}$$

The Wilson Action

The Wilson action is the standard gauge-invariant action for pure gauge theory on the lattice. To see how it can be constructed from plaquettes, one should look at Eq. (22) and recognize that an expansion of the exponential to third order gives us the term appearing in the continuum action in Eq. (1),

$$\frac{1}{e^2} \sum_n \sum_{\mu<\nu} \left[1 - \frac{1}{2}(U_{\mu\nu}(n) + U^\dagger_{\mu\nu}(n)) \right] \approx \frac{1}{4} \sum_{n,\mu,\nu} a^4 F_{\mu\nu}(n) F_{\mu\nu}(n) \,, \tag{26}$$

where the term $1/e^2$, in lattice nomenclature is usually called the *inverse coupling* denoted by β. Also notice that that on the right-hand-side of Eq. (26), an extra

[2] Thanks to Nico Dichter for the OBC results.

factor of $\frac{1}{2}$ appears because the sum over directions now includes both positive and negative directions. Hence, the Wilson gauge action for abelian gauge theory is,

There is an exercise in the second sheet related to the lattice Wilson action and it's naive continuum limit.

$$S_{\mathrm{W}}[U] = \beta \sum_{n} \sum_{\mu<\nu} \left[1 - \frac{1}{2}(U_{\mu\nu}(n) + U^{\dagger}_{\mu\nu}(n))\right] . \tag{27}$$

Typically in simulations, the constant volume term is dropped.

Although we did not derive the corresponding expressions for non-abelian gauge fields like $SU(N)$ in this lecture, it is worth noting that the form of the Wilson action, plaquettes terms remains similar with one important difference being that the links are not phases anymore but $N \times N$ matrices for $SU(N)$ and to ensure gauge invariance, just a product of links over a closed loop is not sufficient - we must take a trace over the plaquette. You are encouraged to go through these calculations using as references [1, 3, 2].

The next exercise sheet also includes the treatment of non-abelian gauge theories.

5 The Lattice Path Integral for $U(1)$ Gauge Theory

Now that we have formulated the lattice abelian gauge theory completely in terms of compact variables like links and plaquettes, we can write down the path integral for the pure gauge theory,

$$Z = \int \mathcal{D}U \; e^{-S_{\mathrm{W}}[U]} , \tag{28}$$

where the path integral measure is given by $\mathcal{D}U \equiv \prod_{n,\mu} dU_{\mu}(n)$.

In order to understand the properties of this measure, we begin by demanding that the path integral Z is a gauge invariant object. One can either argue this by remembering that physical quantities should be gauge invariant or by viewing gauge transformation as change of variables and recalling that integrals are invariant under change of variables. Since we know that the action is gauge invariant by construction, $S_{\mathrm{W}}[U] = S_{\mathrm{W}}[U']$, (with $U' \equiv U'_{\mu}(n) = \Omega(n)U_{\mu}(n)\Omega^{\dagger}(n+\hat{\mu})$), we require the path inetgral measure to be gauge invariant as well. This implies,

$$dU_{\mu}(n) = dU'_{\mu}(n) = d(\Omega(n)\, dU_{\mu}(n)\, \Omega^{\dagger}(n+\hat{\mu})) . \tag{29}$$

An important aspect about gauge transformations is that one can *independently* choose a gauge trasnformation at each point in spacetime (or on the lattice). This gives a stronger condition on the invariance properties of the measure over the measure dU, i.e., the group element must be independently left and right multiplication invariant,

$$dU = d(VU) = d(UV) . \tag{30}$$

One aspect that remains to be understood is: what is the volume associated with this measure? This requires a little discussion because this aspect is fundamentally different in the link formulation from what we encountered in Section 2. Restricting ourselves to formulating lattice gauge theories on links, restricts the integration domain of the vector potentials $aA_{\mu} \in \{-\pi, \pi\}$ (recall that $A_{\mu}(n)$ has dimension $\frac{1}{a}$). This gives us a normalization condition for the measure,

Apparently, this was first introduced by A. Hurwitz in 1897 under the name "invariant integral" (source: Wikipedia)

$$\int dU_{\mu}(n) = \int_{-\pi}^{\pi} \frac{d(aA_{\mu}(n))}{2\pi} = 1 . \tag{31}$$

These conditions, with Eqs. (30) and (31), taken together, define the so-called *Haar measure*. This is a measure for integration over continuous compact groups. This completes our discussion on putting an abelian gauge theory on a lattice while preserving gauge invariance exactly.

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6 Gauge Fixing and Maximal Trees

As we have seen, using the link-formulation, we end up with a well defined and gauge invariant path integral formulation. One can now compute the vacuum expectation values of observables. That is to say we don't have to fix the gauge to perform any computation on the lattice. However, gauge fixing on the lattice can still have some benefits like reducing the dimensionality of the integrated links in Eq. (28) or to make the physical interpretation of some observables more explicit.

Also known as the *compact* formulation.

There is a theorem for gauge theories called **Elitzur's theorem** [7] which essentially states that the vacuum expectation value for non-gauge-invariant observables vanish.

In order to see this we need to understand what gauge-fixing on the lattice means. It turns out that this is quite different to the gauge fixing one is used to in the continuum. To understand this, let us look at the gauge transformation property of the links again,

$$U_\mu(n) \to \Omega(n) U_\mu(n) \Omega^\dagger(n+\mu)\,.$$

This means that the gauge fixing matrices $\Omega(n)$ live at the sites of the lattice. The default values of these matrices is identity. This means we can set these transformations to make some of the links identity - and those links do not participate in the computation of the path integral or any expectation values. For example, consider the Fig. 4, the zigzag lines here represent those links that have been set to identity by using all of the available gauge freedom. This means that no more of the links can be set to identity by any other transformation without making a link that was already identity - not identity anymore.

As an example on how this procedure is performed with respect to Fig. 4, we will write down the transformations that set some of the links to identity. For this we temporarily use the notation $U_n(x)\,, U_n(y)$ to represent the x and y links emanating from site n labelling the lattice sites in the figure. To set $U_1(x) \to \mathbb{I}$, we set $\Omega_1 = U_1^{-1}(x)$ and keep $\Omega_2 = \mathbb{I}$.

Pay attention to new (temporary) notation here !!

$$\begin{aligned} U_1(x) &\to \Omega_1 U_1(x) \Omega_2^\dagger \\ &= U_1^{-1}(x) U_1(x) \mathbb{I} \\ &= \mathbb{I} \end{aligned}$$

Now if we try to use Ω_2 to set the next links, either $U_2(x)$ or $U_2(y)$ to identity, we will run into a problem because $U_1(x)$ will transform away from identity! Hence in order to set the links $U_2(x)$ and $U_2(y)$ to identity we have to use the gauge freedom at sites $n = 3$ and $n = 5$ respectively. The tranformations that set $U_2(x)$ and $U_2(y)$ to identity are $\Omega_3 = U_2(x)$ and $\Omega_5 = U_2(y)$. As an example of a non-trivial tranformation in Fig. 4, consider the link $U_6(x)$. Since $\Omega_5 = U_2(y)$ has already been

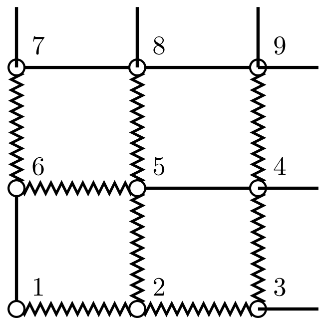

Figure 4: An example of a maximal tree on a 3×3 lattice with periodic boundary conditions.

used, we need to use Ω_6 to set it to identity. The transformations are as follows,

$$\begin{aligned} U_6(x) &\rightarrow \Omega_6\, U_6(x)\, \Omega_5^\dagger \\ &= \Omega_6\, U_6(x)\, U_2^{-1}(y) \\ &= \Big(U_2(y)U_6^{-1}(x)\Big)\, U_6(x)\, U_2^{-1}(y) \\ &= \mathbb{I}\,, \end{aligned}$$

which shows that $\Omega_6 = U_2(y)U_6^{-1}(x)$ was needed to transform $U_6(x) \rightarrow \mathbb{I}$. One can similarly set the remaining gauge transformation matrices at other sites to set the zigzag links shown in the figure to identity. An important point to note is that the maximal tree structure is *not unique*!

7 Important Observables: Wilson and Polyakov Loops

When considering gauge theories, there are two observables of high importance: the Wilson loop and the Polyakov loop. These are only briefly discussed below with a focus on what physical interpretation they have - which becomes transparent when we fix the gauge by using the maximal tree construction above. For a detailed discussion you are referred to the book by Gattringer & Lang [3].

Wilson Loop

Following the notation of ref. [3], we can define the Wilson loop as a product over four link segments: two Wilson lines $S(m,n,0)$, $S(m,n,n_t)$ and two temporal transporters $T(n,n_t)$ and $T(m,n_t)$,

A trace over the product is required for non-abelian gauge fields.

$$W_L = S(m,n,0)T(n,n_t)S^\dagger(m,n,n_t)T^\dagger(m,n_t)\,. \tag{32}$$

Here the Wilson lines are formed by a product over spatial links at a fixed time slice, while the temporal transporters are formed by a product over time-like links at a fixed spatial position. In order to see the physical significance of the Wilson loop one usually considers the *temporal* gauge, in which time-like links are set to the identity, making $T(n,n_t)$ and $T(m,n_t)$ trivial. This leaves the Wilson loop as a product of two Wilson lines, and its expectation value,

$$\langle W_L \rangle = \langle S(m,n,0)S^\dagger(m,n,n_t) \rangle\,, \tag{33}$$

can be interpreted as a correlator of two Wilson lines. The large-time behaviour of such correlators encodes the low-lying spectrum of the theory - in the case of non-abelian gauge theory, this gives us access to the static quark-antiquark potential.

Polyakov Loop

Once again, following the notation of ref. [3], the Polyakov loop is defined as a product of temporal links at a fixed spatial position n, wrapping around the entire temporal extent of the lattice,

$$P(n) = \prod_{n_t=0}^{N_t-1} U_4(n, n_t)\,. \tag{34}$$

Unlike the Wilson loop, the Polyakov loop is a non-contractible loop that closes only due to the periodic temporal boundary conditions. Its expectation value is related to the free energy F_q of a single static quark, and serves as an order parameter for the deconfinement phase transition. The correlator of two Polyakov loops at spatial separation r,

Here non-contractible means that the loop cannot be continuously shrunk to a point.

$$\langle P(0)P^\dagger(r)\rangle \sim e^{-N_t a\, F(r)}\,, \tag{35}$$

encodes the free energy of a static quark-antiquark pair at finite temperature, and can be seen as the finite-temperature analogue of the static potential extracted from the Wilson loop.

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